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UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

COURSE CODE : SECD2613
COURSE NAME : SYSTEM ANALYSIS AND DESIGN
YEAR / PROGRAM : 1 / 2 (SECB/SECJ/SECR/SECV)
TIME : 07 FEBRUARY 2023
DATE :
PLACE :

INSTRUCTIONS TO CANDIDATES:

1. Fill in your particulars in the question booklet and answering booklet:

Name	AKMAL RAFLQVE BIN AHMAD RAHMAIE
Passport No.	
Matric No.	A25080181
Year / Program	1/SECB
Lecturer	

2. Read instructions given for each question carefully.
3. Answer **ALL** questions.
4. This paper consists of three sections.

Section A: **10 MULTIPLE CHOICE** questions (10 Marks).

Section B: **4 STRUCTURED** questions (60 Marks).

Section C: **1 CASE STUDY** question (30 Marks).

TOTAL MARKS is **100 Marks**

THIS PAPER CONSISTS OF 13 (THIRTEEN) PRINTED PAGES INCLUDING THIS PAGE.

SECTION A : MULTIPLE CHOICE QUESTIONS
10 MARKS (EACH QUESTION 1 MARK)

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer (A/B/C/D) in the answer sheets.

Question 1

Which of the statements **BEST** describe a logical data flow diagram?

- A. Logical data flow diagram includes types of programs, such as online or batch.
- B. Logical data flow diagram is a model of how the system will be implemented.
- C. Logical data flow diagram does not include any primitive processes.
- ☒ D. Logical data flow diagram is a model of how the business operates.

Question 2

X is use when not every condition is relevant to every action.

Based on the above statement, what is X?

- A. Structured English
- B. Decision Table
- C. Structured Tree
- ☒ D. Decision Tree

Question 3

Which of the following is the purpose of a control couple, in a structure chart?

- ☒ A. Pass parameters from a subordinate module to the control module.
- B. Pass parameters from the control modules to a subordinate module.
- C. Pass data from a subordinate module to the control module.
- D. Pass data from the control module to a subordinate module.

Question 4

Which of the following is **NOT** a form of electronic output?

- A. Fax
- B. Email
- ☒ C. Report
- D. Bulletin Board Message

Question 5

Which of the following is a **FALSE** statement regarding the purpose of system implementation?

- A. To convert final physical system specifications into working and reliable software.
- B. To document work that has been done.
- C. To provide help and support for current and future users.
- ☒ D. To modify a system to correct faults, to improve performance, or to adapt the system to a changed environment or changed requirements

Question 6

Which of the following is **FALSE** about the user interface design?

- A. Availability of operating systems will be supported.
- ☒ B. Minimize the usability and interaction between devices and users.
- C. Types of devices will be used.
- D. The system is designed to use a browser.

should be maximize

Question 7

Which of the following is **FALSE** about the decision table?

- A. Effectively avoids impossible situations ✓
- B. Effectively avoids redundancies ✓
- ☒ C. Effectively avoids miscommunication to end users
- D. Effectively avoids contradictions ✓

Question 8

Consider the transaction below:

Transaction 1: add new staff

Transaction 2: change a venue of an event

Transaction 3: change a customer phone number

Transaction 4: show a list of events in 2020

Transaction 5: change a staff bank account number

Which of the following CRUD matrix will represent the above transaction?

A.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		U	
Transaction 3	C		
Transaction 4		R	
Transaction 5			D

B.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		C	
Transaction 3	C		
Transaction 4			R
Transaction 5			D

C.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		U	
Transaction 3	U		
Transaction 4		R	
Transaction 5			U

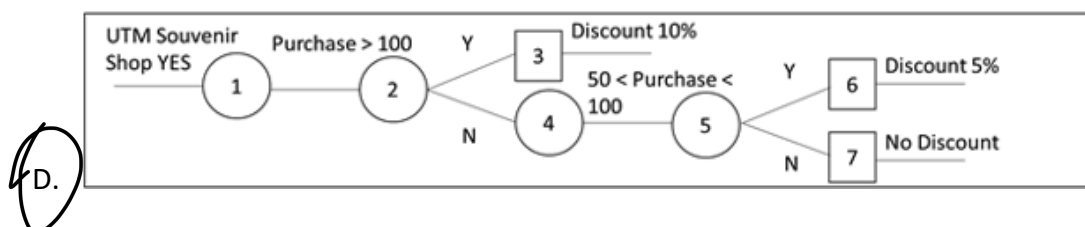
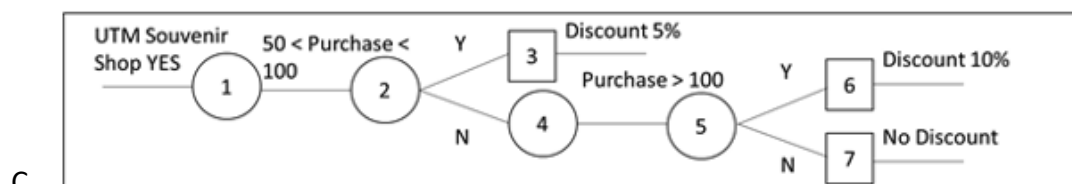
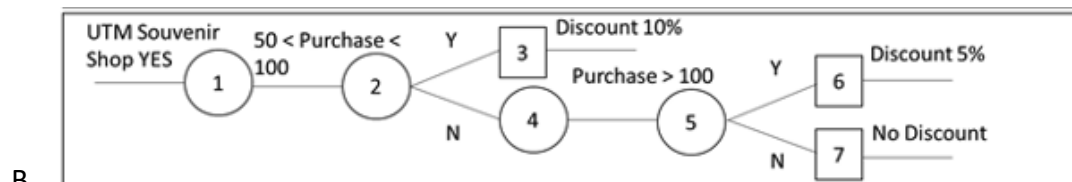
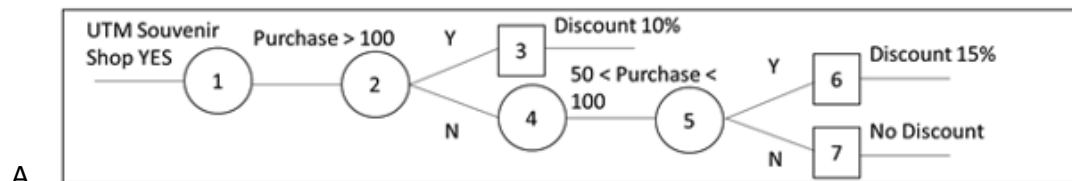
D.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		C	
Transaction 3	U		
Transaction 4		R	R
Transaction 5			D

Question 9

UTM Souvenir Shop plan to conduct Year End Sale, where if the amount of the purchase is greater than RM100, the customer will received 10% discount, if the amount of the purchase is between RM50 and RM100, the customer will be received 5% discount and if the amount is less than RM 50, the customer will not receive any discount.

Which one of the decision trees below is **TRUE** to represent the above scenario?



Question 10

XYZ University decided to change the registration information system and to replace it with a new system. The new registration system was developed internally by the information technology staff. Now, the information technology staff want to install the new registration system. The manager of the registration department requires that the installation involves comparing the results of the new system with the old one.

Based on the above scenario, which of the following installation strategies is the **MOST** appropriate?

- A. Direct Installation
- ☒ B. Parallel Installation
- C. Single-location Installation
- D. Phased Installation

SECTION B : STRUCTURED QUESTIONS

60 MARKS

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer in the answer sheets.

QUESTION 1 (15 Marks)

Figure 1 shows Logical DFD Level 1 for one of the processes in Cinematics Ticketing System.

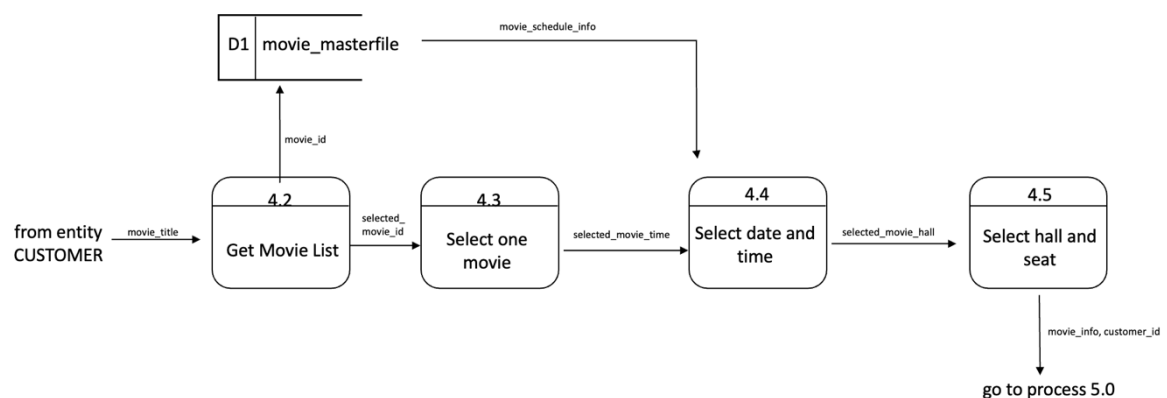


Figure 1: Logical DFD Level 1 (Process 4.0 CHOOSE MOVIE)

A detailed flow for *Process 4.0 Choose movie* is explained as below:

The process starts when the customer goes to the movie section on the website. The list is retrieved from the movie_masterfile and a complete list was shown to the customer. Based on the list, the customer can choose one movie that they are interested in. The selected movie will get the schedule data from a temporary schedule file and the customer is allowed to choose the date and time that they preferred. Then the hall that has the movie show that was selected will be displayed so the customer can select the seating that they interested. The hall and seating view will also display the ticket price. Note that different seating would have different price, and this is kept in another temporary file for the ticket price. After the seat was chosen, the total price is displayed, and the customer will be taken to the payment page (Process 5.0). This is the end for Process 4.0.

- a) Identify the main processes from the description that can assist you to convert Figure 1 to a PHYSICAL DFD.

(Hint: It can be based on Figure 1 and other process you can identified from the description)

[6 Marks]

- b) Draw a PHYSICAL DFD Level 1 for *Process 4.0 Choose movie* based on your answer in (a).

[9 Marks]

QUESTION 2 (15 Marks)

- a) What is a structure chart, and why is it useful? **(2 Marks)**

- b) Figure 2 describes a new order process for the Food Ordering System. Convert the DFD in Figure 2 into a Structure Chart. **(7 Marks)**

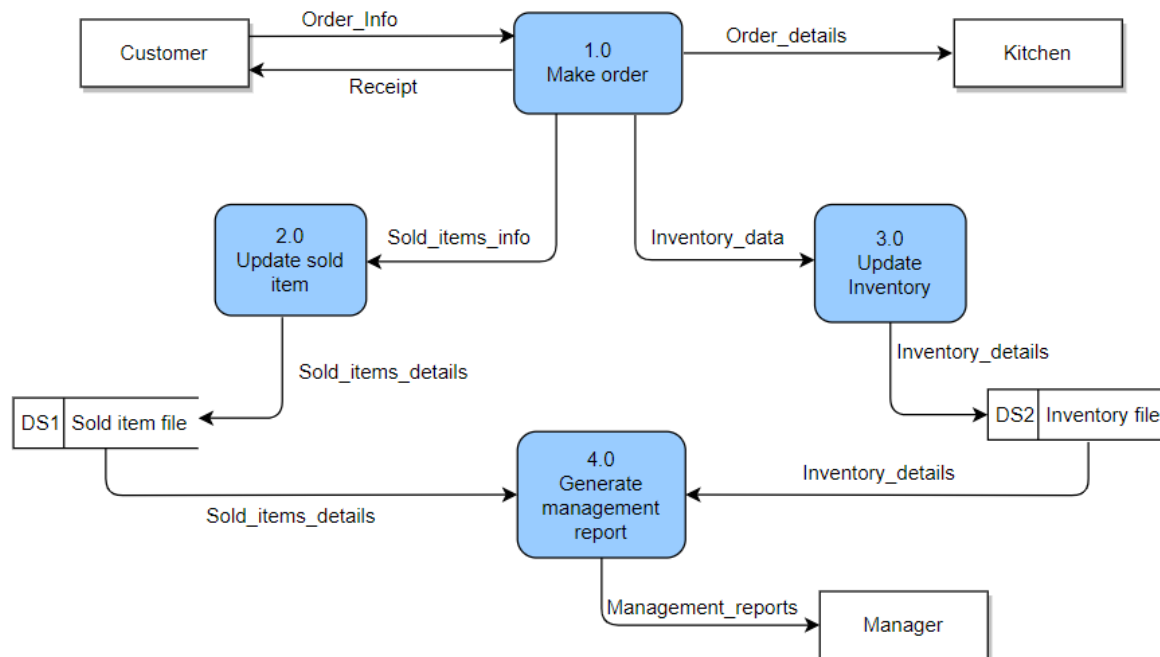


Figure 2: DFD for the Food Ordering System

- c) Based on your answer in (b), identify a module which has a high degree of cohesion and is loosely coupled. Provide your justification. **(2 Marks)**

- a) A structure chart, or a functional decomposition diagram is a top down representation of business functions and process. It is useful because it shows the program modules and the relationship among them.
- c) Module 2, The module perform a single function, which is sending sold item details into the sold item file data store, requires only one input. not affecting other modules and is an independent module.
- d) START Manage order process
Repeat
call collect order
call verify order
call make payment
END Repeat
call generate order list
END START

```
START Program
DoUntil No more time cards
    Call Enter time cards
    Call Calculate amounts
    Call Output payroll
END Until
END START
```

d) Write the Structured English for the structure chart in Figure 3.

(4 Marks)

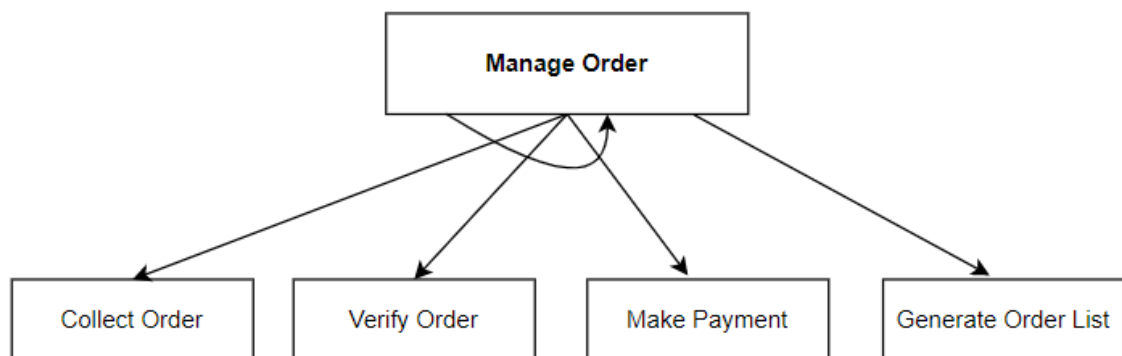


Figure 3: Structured chart for Manage Order Process

QUESTION 3 (15 Marks)

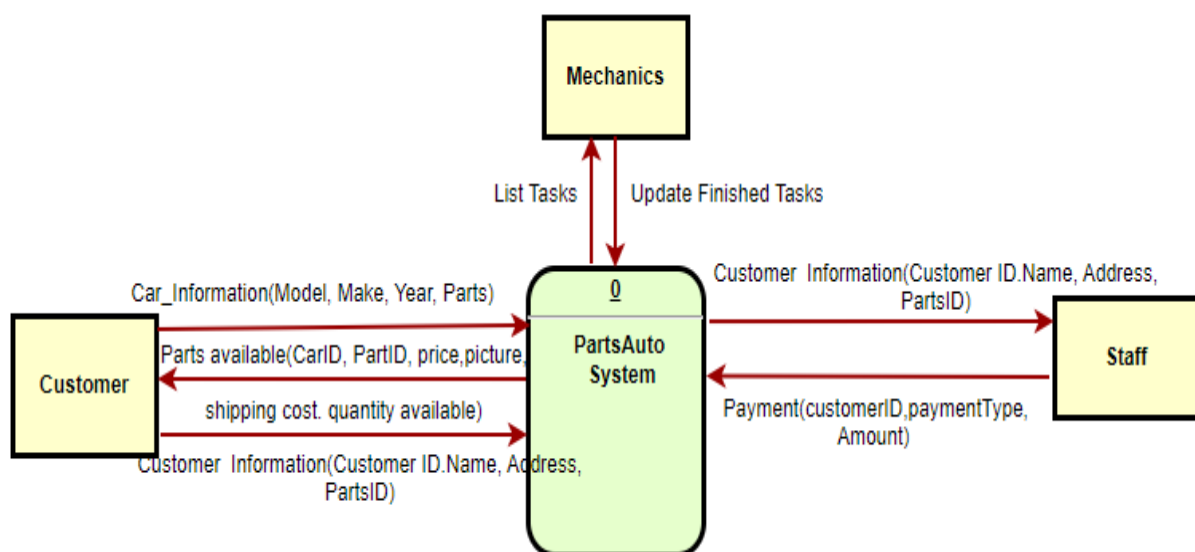


Figure 4: Context Diagram

PartsAuto is a company selling recycling parts for classic and antique cars. Customer can search for information related to their car and find available parts for sale. Mechanics can be assigned task for installing the parts in the customer's car and update their finished task while staff will manage the customers history of purchased parts and their payments.

- a) Based on the Context Diagram shown in Figure 4 identify **THREE** input and **THREE** output screens to be designed. **(6 Marks)**
- b) You have been given the task to design a Web-based display for Parts Auto Company where:
- Customer can search for make, model, and year. When the data of the car is found, it will show all the available parts of the car.
 - Information for the car parts are: the stock is in or not; the part description; condition of the part; price, and shipping cost are displayed; how many quantity available for each part, along with a picture of the part, button for each part that may be clicked to purchase the part.

Figure 5 shows the initial design which really needs improvements. Improve the GUI design by drawing the GUI with suitable text boxes, varieties of GUI such as drop-down menus, check boxes, and radio buttons which are suitable to create interest in the form. **(9 Marks)**

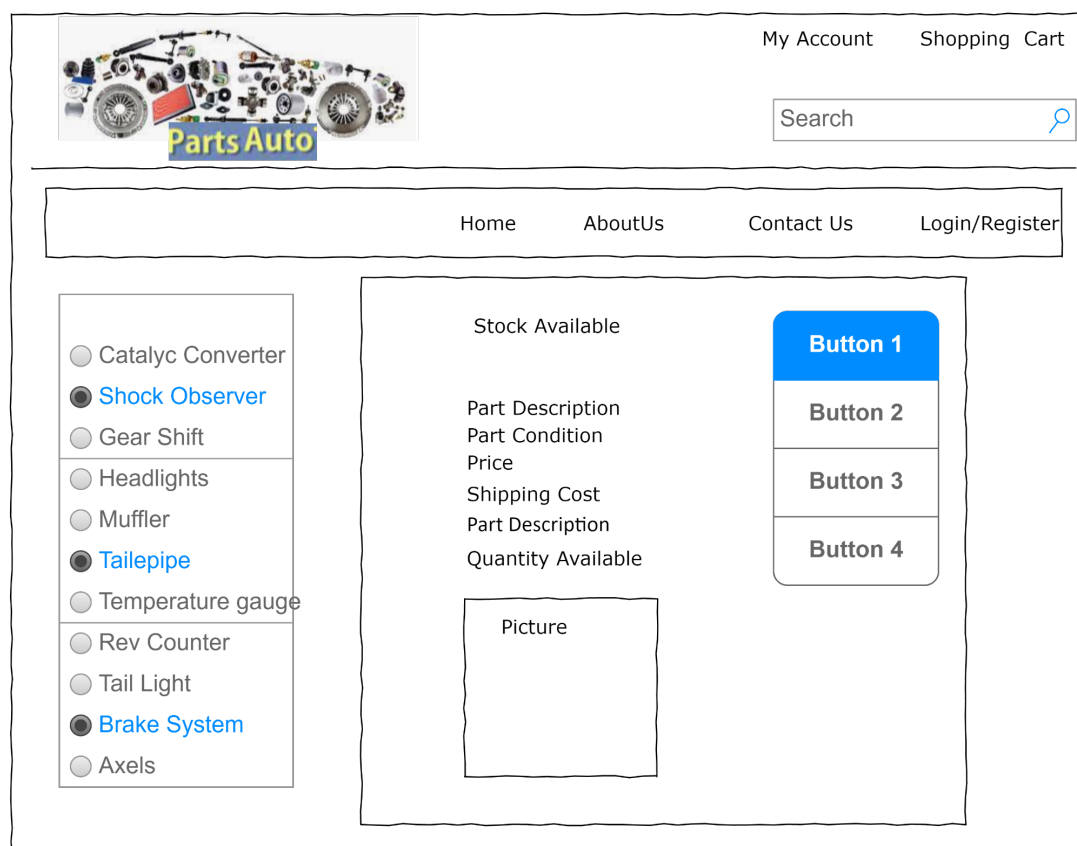


Figure 5: Initial GUI Design

a) 1. Car information and customer information screen for customer

~~2. Update finished task screen for mechanics~~

3. Payment processing screen

4. mechanic task update screen

Output screen

1. Parts available screen for customer

2. list task for mechanics

3. Customer information display screen for staff
 ↳ purchase history screen

1/2

QUESTION 4 (15 Marks)

MyTown Sdn. Bhd. developed a new e-commerce system called MyTown.MY that will replace their existing system, MiTown.MY. As the system analyst in that company, you need to ensure that the system implementation process is well conducted before the company launches that new system.

- a) Firstly, you need to give a briefing to your top management about the system implementation process. List **THREE** (3) process of systems implementation. **(3 Marks)**
- b) Next, top management requests you to conduct an acceptance test before the company launches that new system. Describe **THREE** (3) types of acceptance testing that you can propose to them. **(6 Marks)**
- c) Finally, below are **THREE** (3) scenarios given by your top management related to installation strategy for the new system. Identify the most suitable installation strategy for each scenario and justify the **ONE** (1) advantage of using the strategy. **(6 Marks)**
 - i. The top management plans to run both systems at the same time. *para*
 - ii. The top management plans to change the old system to the new one incrementally. *phases*
 - iii. The top management plans to change over from the old system to a new one immediately. *direct*

- a)
- 1) coding
 - 2) testing
 - 3) installation



b) 1) user acceptance testing → this is conducted by or visible to the end users and customer.
to ensure the system meets the user requirement

2) Operational acceptance testing → acceptance of the system by the personnel who have administered it. It covers features such as testing of backup and restore procedures and disaster recovery plan to ensure the system can be supported

3) Regulation acceptance testing → This testing is performed against any regulations which must be followed such as governmental, legal or safety regulation.

c) 1) parallel installation → run both systems at the same time and decide to turn off the old system when decided.
- can compare result of new and old one.
- offers low risk because the old system remains operational as backup

phased installation → provide user learning advantage because employees and customers can become familiar with specific functional components of new system one at a time.

direct installation → There is no duplication of effort by the users because they do not have to perform the same task in both the old and new system simultaneously.

SECTION C : CASE STUDY

30 MARKS

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer in the answer sheets.

In early 2019, GigaBites - a computer supplies company has set up number of new branches for their business operation in Johor. GigaBites sends out invoices monthly and will offers discounts if the clients (made payments within 18 days after purchases.) Due to COVID-19 pandemic that disrupts the economy globally, GigaBites decides to revise the discounting policy for the clients to sustain their business environment.

The revised discounting policy is as follows: If the amount of the order for (computer supplies is greater than \$1,000,) subtract 4 percent for the order; If the amount is (between \$500 and \$1,000,) subtract a 2 percent discount; If the amount is (less than \$500,) do not apply any discount; All orders made via the Web automatically receive an extra 5 percent discount; (Any special goods order (computer tables, for example)) is exempt from all discounting.

- a) Propose the structured English to solve the GigaBites company situation for the revised discounting policy. **(6 Marks)**
- b) Develop a simplified decision table for GigaBites discounting decisions, for which the condition alternatives are limited to Y and N. **(8 Marks)**
- c) Draw a decision tree to illustrate the possible outcomes for the revised discounting decisions. **(6 Marks)**
- d) Sketch the graphical user interface (GUI) design for the GigaBites company manager to overview the sales summary report using their inventory management system. **(8 Marks)**

The GUI design should fulfill the following needs highlighted by the manager during interview:

"This report should give me an (overview of GigaBites sales for a given time period) The (generated sales report should also show the cost of goods sold, gross profit, margins, and

a)

BEGIN discount

If order is special goods
apply no discount

ELSE

IF amount > \$1000
APPLY 4% discount

ELSE IF amount ≥ 500 & < 1000
APPLY 2% discount

ELSE
APPLY 0% discount

END IF

END IF

IF order via web

APPLY extra 5% discount

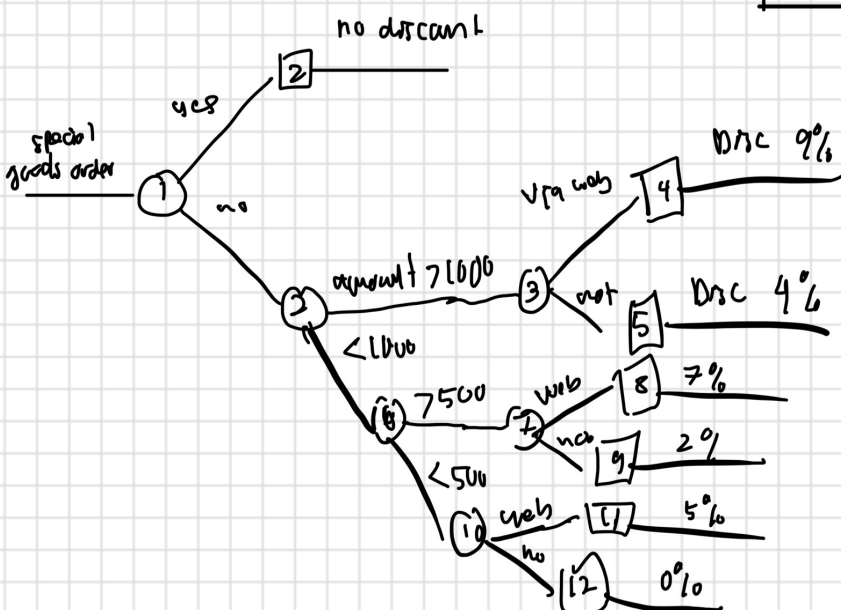
END IF

SEND final amount to customer

END discount

Condition	Rules
	1 2 3 4 5 6 7
Special goods order	Y N N N N N N
amount > 1000	- Y N N Y N N
amount 500-1000	- N Y Y N Y N
amount < 500	- N N Y N N Y
order via web	- N N N Y Y Y
Action	
Apply 4% d/s	X X
2%	X
0%	X X X
extra 5% discount	X X X

c)



GIGABYTE IMPORT/EXPORT MANAGEMENT SYSTEM		
Sales Summary report		
Select month	APRIL	
Date:	1/4/26	to 31/4/26
Financial overview:		
total sales by time period \$67,000		
Financial details:		
goods sold	\$10,000	TAX
gross profit	\$10,000	goods sold
margin	46%	\$10,000
metrics:		
last month:	1000	net rev
		20000
		profit
		+10,000

tax, so I can have a better idea of how much money is actually going back to GigaBites business.

Also, the sales summary report provides a more macro view of GigaBites retail sales: How much did company make last month or the previous month? What about company year-to-date sales compared to last year? These numbers should provide some general insights around the health of GigaBites business, and it can inform GigaBites medium to long-term decisions."

- e) Justify the use of appropriate output elements that are included in the GUI design in Question d. (2 Marks)

Output element includes table which are clear and easy to read. This ease the manager to make quick and efficient decision making